

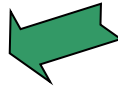
ED Hospital: Process Discovery Results

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Agenda

- **Exploratory Analysis**
- Time Perspective
- Longest Variants
- Infrequent Variants
- Alpha+ Miner BP Model
- Inductive Miner BP Model
- Heuristic Miner BP Model
- Our Team's BP Model
- Insights





Exploratory Analysis on Event Log Using PM4PY

Most Common Variant (count: 7994)

1. Reception
2. Nurse Admission
3. Order Blood Tests
4. Last Lab Tests Results
5. Doctor Admission
6. Discharge (Doctor)
7. Discharge

The Most Common Activity (count: 55079)

- The most common activities are tied at Reception and Discharge

Activity	
Reception	55079
Discharge	55079
Doctor admission	54871
Nurse admission	54300
Discharge (Doctor)	54213
Last lab tests results	41268
Order Imaging test	29730
Order blood tests	28043
Imaging	27824
Additional vitals	25188
Order Consultant	15192
Consultant	12954
Imaging decrypting	7110
Order external exams	506
External exams	482
Name: count, dtype: int64	



Time Perspective

Activity	Average Flow Time	Average Delays in Flow Time
Nurse admission	797.75	3075.17
Doctor admission	1047.51	3984.34
Order blood tests	1456.76	1192.77
Reception	1965.83	1496.97
Order imaging test	2067.35	3938.95
Last lab test results	2971.52	4910.96
Order consultant	3126.36	5273.54



Time Perspective Cont'd

Activity	Average Flow Time	Average Delays in Flow Time
Additional vitals	3968.83	10150.99
Discharge (Doctor)	4701.74	8510.92
Imaging	4961.36	8196.99
Imaging decrypting	5714.28	12764.49
Consultant	5797.12	15756.64
External exams	6578.72	26068.35
Order external exams	10575.06	16784.71



Longest Variant In Terms of Flow Time

Variant	('Reception', 'Last lab tests results', 'Order Consultant', 'Consultant', 'External exams', 'External exams', 'Order Consultant', 'Consultant', 'Order Consultant', 'Order Consultant', 'Order Consultant', 'Consultant', 'Order Consultant', 'External exams', 'Consultant', 'Order Consultant', 'Consultant', 'Order Consultant', 'Consultant', 'Order Consultant', 'Consultant', 'Consultant', 'Order Consultant', 'Consultant', 'Nurse admission', 'Additional vitals', 'Doctor admission', 'Additional vitals', 'Discharge (Doctor)', 'Discharge')
Total Mean Flow Time	4826340 seconds ~ 55 days 20:39:00
# of Activities	30



Longest Variant In Terms of # of Activities

Variant	('Reception', 'Nurse admission', 'Order blood tests', 'Last lab tests results', 'Doctor admission', 'Order Consultant', 'Order Consultant', 'Order Imaging test', 'Consultant', 'Imaging', 'Order Imaging test', 'Imaging', 'Imaging decrypting', 'Order Imaging test', 'Order Consultant', 'Consultant', 'Order Consultant', 'Order Consultant', 'Imaging decrypting', 'Consultant', 'Order Imaging test', 'Imaging', 'Imaging decrypting', 'Order Consultant', 'Order Consultant', 'Imaging', 'Order Consultant', 'Order Consultant', 'Consultant', 'Consultant', 'Consultant', 'Imaging decrypting', 'Discharge (Doctor)', 'Consultant', 'Discharge')
Total Mean Flow Time	89400 seconds ~ 1 day 00:50:00
# of Activities	35

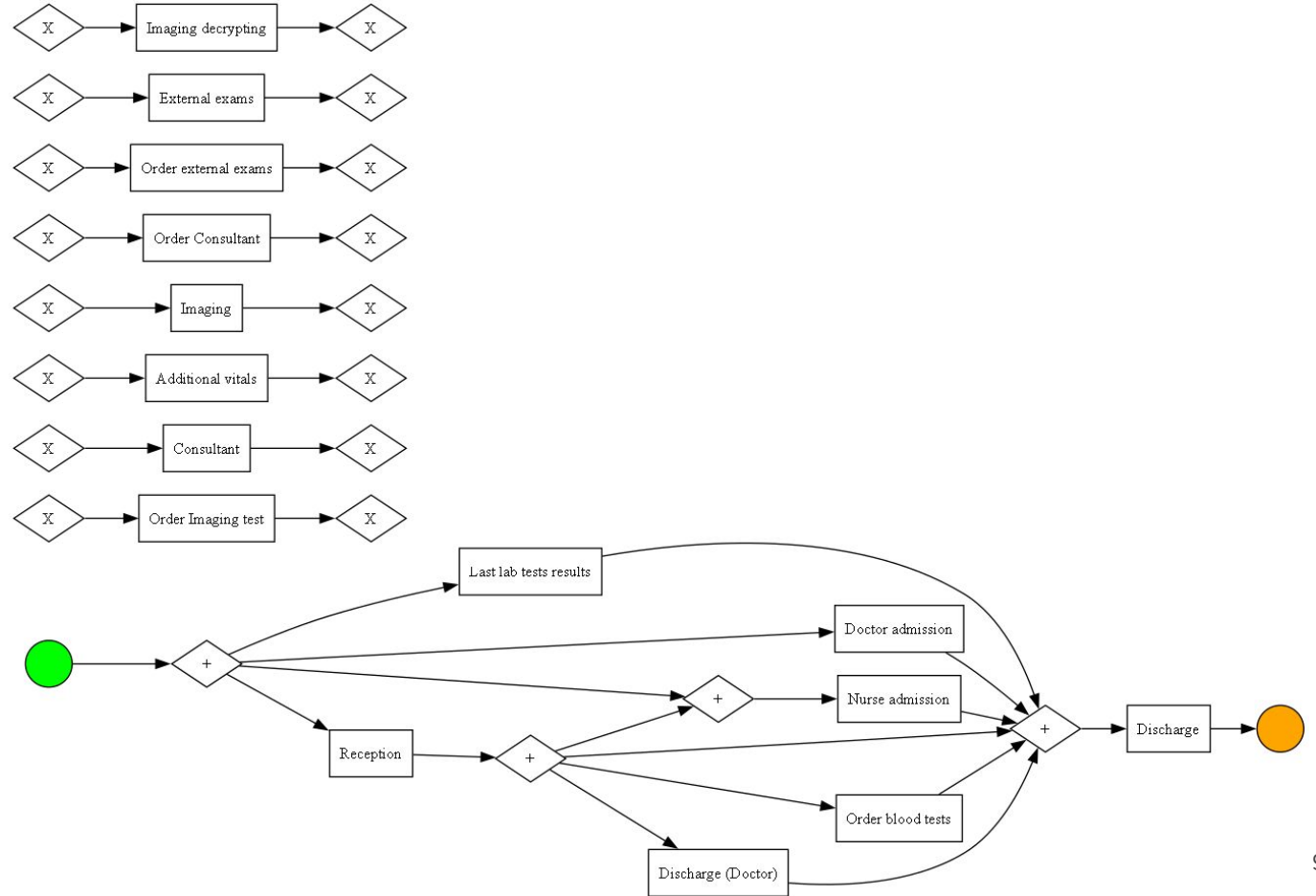


Infrequent Variants

7703 variants occur only once

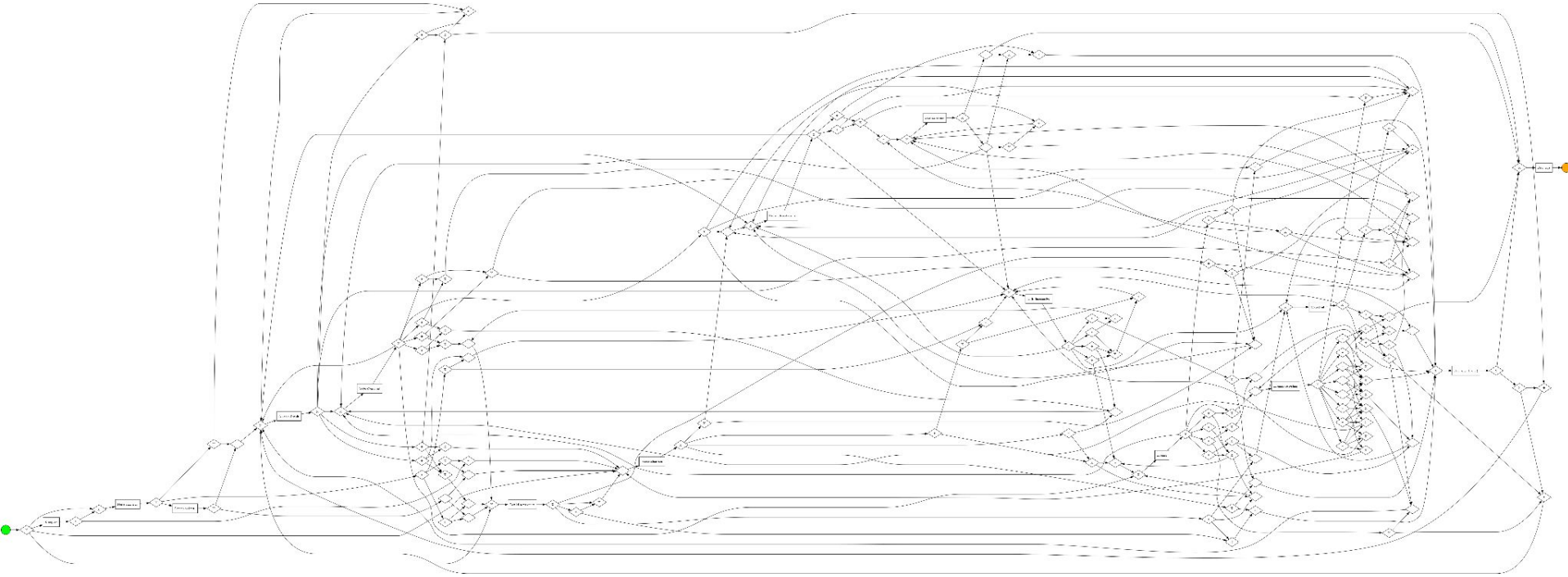
Variants	Average Total Flow Time	Average # of Activities
Infrequent	40455.87	13.41
Frequent	27169.66	10.53

Alpha+ Miner BP Model

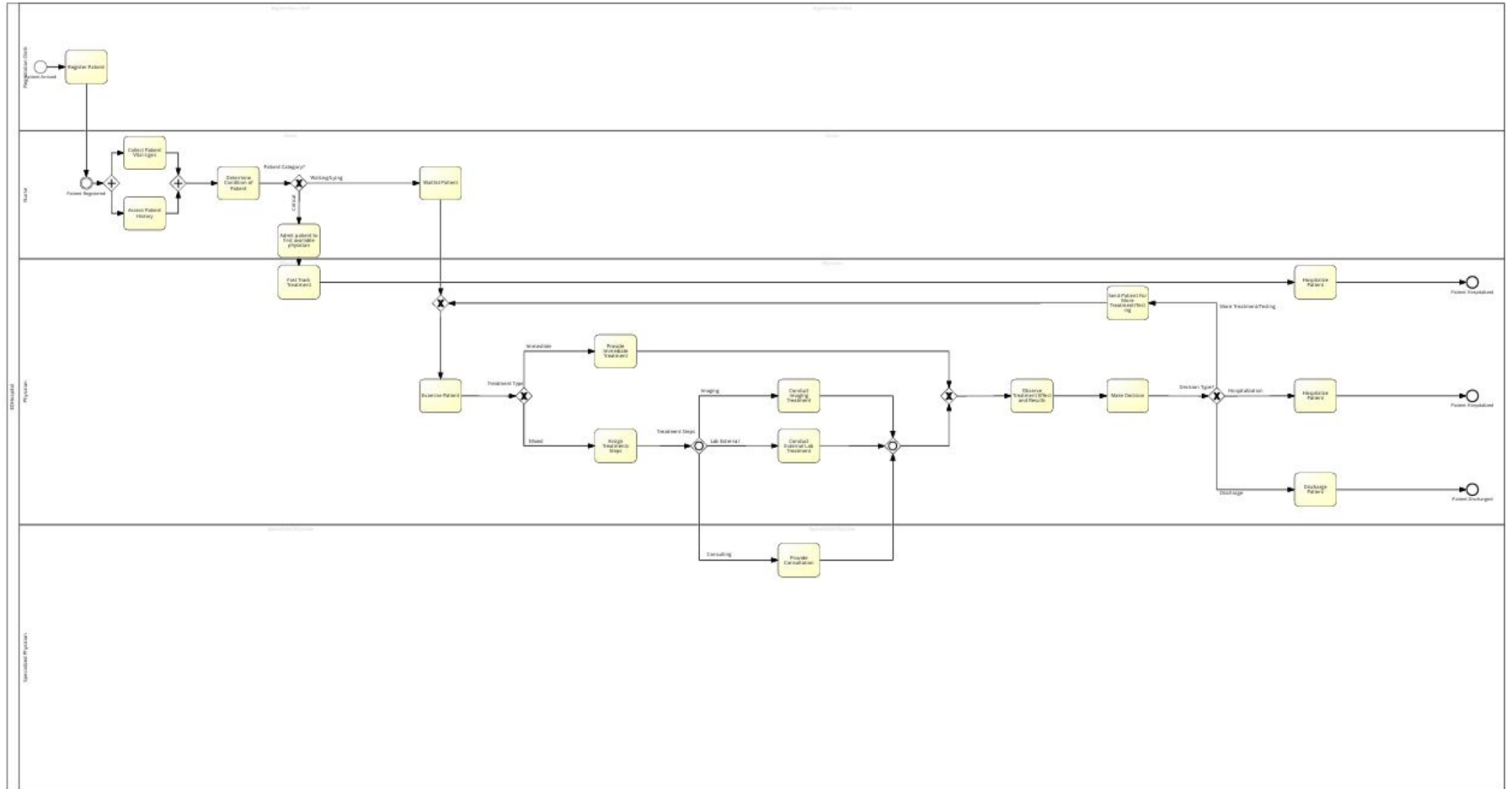




Heuristic Miner BP Model



Our Team's BP Model



Insights

PROCESS MINING ALGORITHMS: A COMPARISON OF GENERATED MODELS

